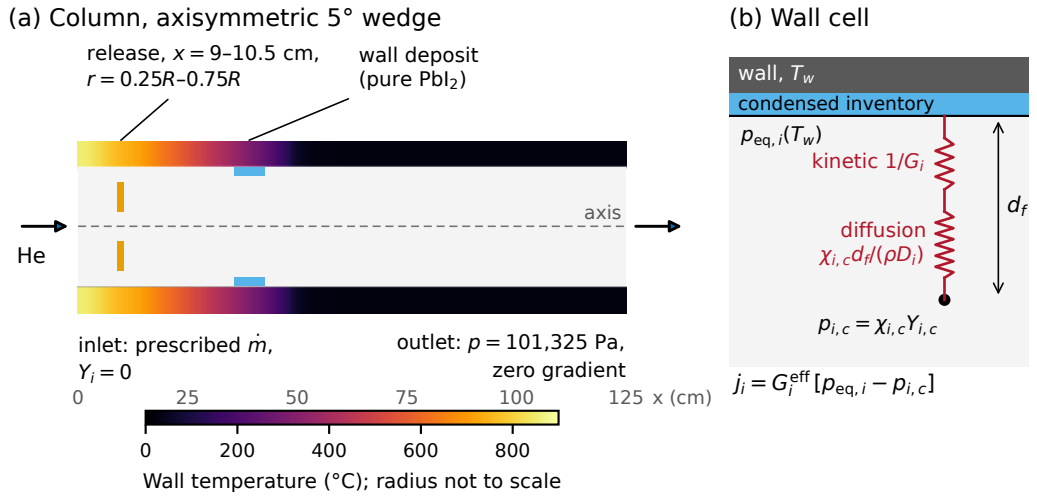
(a) Column, axisymmetric 5° wedge
release, $x = 9$–$10.5$ cm,
$r = 0.25R$–$0.75R$
wall deposit
(pure $PbI_2$)
axis
He
inlet: prescribed $\dot{m}$,
$Y_i = 0$
outlet: $p = 101{,}325$ Pa,
zero gradient
0
25
50
75
100
125 x (cm)
0
200
400
600
800
Wall temperature (°C); radius not to scale
(b) Wall cell
wall, $T_w$
condensed inventory
$p_{eq,i}(T_w)$
kinetic $1/G_i$
diffusion
$\chi_{i,c} d_f/(\rho D_i)$
$d_f$
$p_{i,c} = \chi_{i,c} Y_{i,c}$
$j_i = G_i^{eff}[p_{eq,i} - p_{i,c}]$